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ECE 2040: Homework 4

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Problem 4.5-2

The values of the mesh currents in the circuit shown in Figure P 4.5-2 are

$$i_1 = 2\,\text{A}, \quad i_2 = 3\,\text{A}, \quad i_3 = 4\,\text{A}$$

Determine the values of the resistance R and of the voltages v_1 and v_2 of the voltage sources.

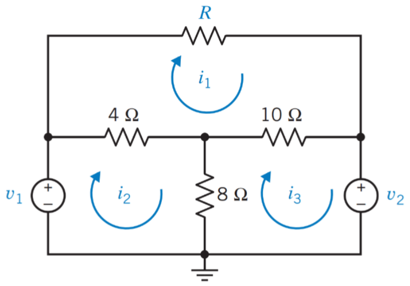

Figure P 4.5-2

$-v_1 + 4(i_2 - i_1) + 8(i_2 - i_3) = 0$

$8(i_3 - i_2) + 10(i_3 - i_1) + v_2 = 0$

$10(i_1 - i_3) + 4(i_1 - i_2) + i_1 R = 0$

$-v_1 + 4 - 8 = 0$ $\quad$ $8(1) + 10(2) + v_2 = 0$

$-v_1 - 4 = 0$ $\quad$ $v_2 = -28\,V$

$v_1 = -4\,V$

$10(-2) + 4(-1) + 2R = 0$

$2R = 24$

$R = 12\,\Omega$

Problem 4.5-4

Determine the mesh currents, i_a and i_b, in the circuit shown in Figure P 4.5-4.

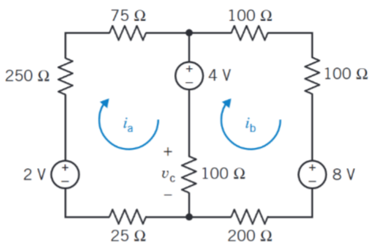

Figure P 4.5-4

$-2 + 250 i_a + 75 i_a + 4 + 100(i_a - i_b) + 25 i_a = 0$

$-4 + 100 i_b + 100 i_b + 8 + 200 i_b + 100(i_b - i_a) = 0$

$2 + 450 i_a - 100 i_b = 0$

$450 i_a - 100 i_b = -2$

$500 i_b - 100 i_a = -4$

$4.5 i_a - i_b = 20\,mA$

$-i_a + 5 i_b = -40\,mA$

Problem 4.5-5

Find the current i for the circuit of Figure P 4.5-5.
Hint: A short circuit can be treated as a 0-V voltage source.

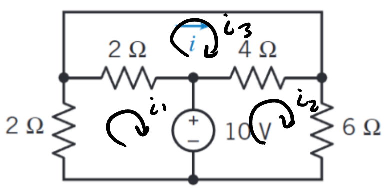

Figure P 4.5-5

$2i_1 + 2(i_1 - i_3) + 10 = 0$

$-10 + 4(i_2 - i_3) + 6 i_2 = 0$

$2(i_3 - i_1) + 4(i_3 - i_2) = 0$

$4i_1 - 2i_3 = -10$ $\quad$ $i = i_3$

$10 i_2 - 4 i_3 = 10$

$-2i_1 - 4i_2 + 6i_3 = 0$

$$\begin{bmatrix} 4 & 0 & -2 \\ 0 & 10 & -4 \\ -2 & -4 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \\ 0 \end{bmatrix}$$

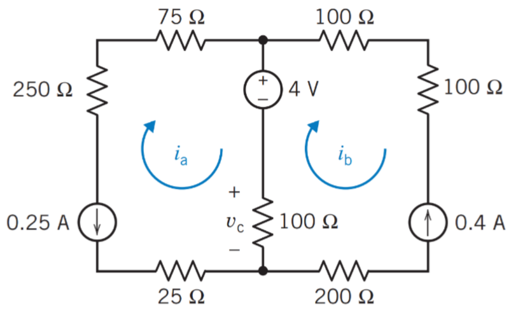

Figure P 4.6-2

$i_a = -0.25\,A$

$i_b = -0.4\,A$

$V_c = (i_a - i_b) \cdot 100\Omega$

$V_c = 15\,V$

Problem 4.6-4

Find v_c for the circuit shown in Figure P 4.6-4.

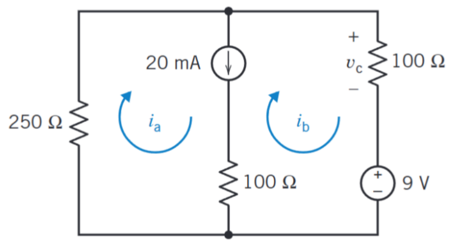

Figure P 4.6-4

$i_a - i_b = 20\,mA$

$250 i_a + 100 i_b + 9 = 0$

$250(20mA + i_b) + 100 i_b = -9$

$5A + 350 i_b = -9A$

$350 i_b = -14A$ $\quad i_b = -40\,mA$ $\quad V_c = 100\,\Omega \cdot -40\,mA$

$i_a = -20\,mA$ $\quad V_c = -4\,V$

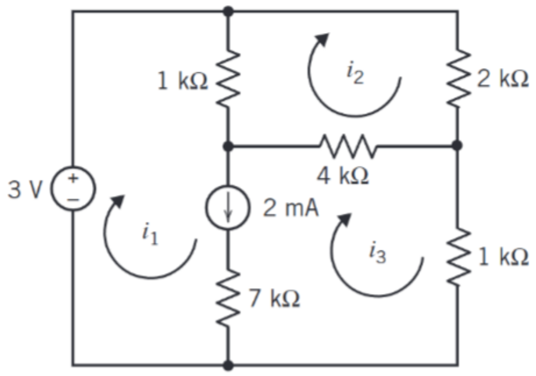

Figure P 4.6-8

$4k\Omega(i_2 - i_3) + 1k\Omega(i_2 - i_1) + 2k\Omega(i_2)$

$7k\Omega\, i_2 - 1k\Omega\, i_1 - 4k\Omega\, i_3 = 0$

$i_1 - i_3 = 2mA$

$-3V + 1k\Omega(i_1 - i_2) + 4k\Omega(i_3 - i_2)$

$+ 1k\Omega(i_3) = 0$

$1k\Omega\, i_1 - 5k\Omega\, i_2 + 5k\Omega\, i_3 = 3V$

Values of i_1, i_2, i_3 will be in mA.

$$\begin{pmatrix} 1 & 0 & -1 \\ 1 & -5 & 5 \\ -1 & 7 & -4 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$$

$\frac{2-V_2}{4k\Omega} = i_a$ $\frac{2-V_2}{4k\Omega} = 0.125mA \rightarrow 2-V_2 = 0.5V$

$V_2 = 1.5V$

$-2V + 4k\Omega\, i_a + 2k\Omega(i_a + 5i_a) = 0$

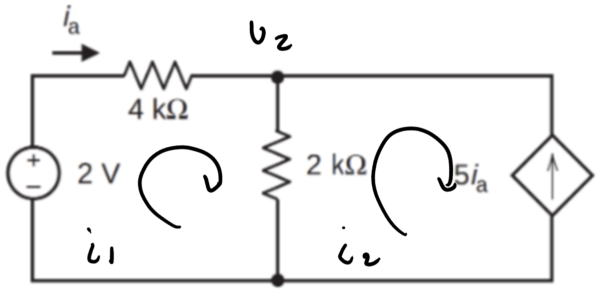

Figure P 4.7-2

$16k\Omega\, i_a = 2V$

$i_a = 0.125mA$

CCCS $= 5i_a = 0.625mA$

$p = vi = 2V \cdot 0.125mA = 0.25mW$

$p = vi = 1.5V \cdot 0.625mA = 0.9375mW$

$i_1 = i_a$

$i_2 = -5i_a$

Problem 4.7-4

Determine the mesh current i_a for the circuit shown in Figure P 4.7-4.

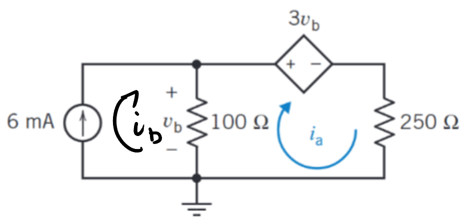

Figure P 4.7-4

$3V_b + 250i_a + 100(i_a - i_b) = 0$

$350i_a - 100i_b = -3V_b$

$100(i_b - i_a) = V_b$

$V_b = 100i_b - 100i_a$

$-3V_b = 350i_a - 100i_b$

$V_b = .6 - 100i_a$

$-3V_b = 350i_a - .6$

$i_b = 6mA$

$-1.8 + 300i_a = 350i_a - .6$

$-1.2 = 50i_a$

$i_a = -24mA$

$V_b = .6 + 2.4 = 3V$

Problem 4.7-9 ★

Determine the value of the resistance R in the circuit shown in Figure P 4.7-9.

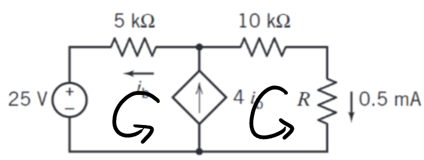

Figure P 4.7-9

$i_a = -0.5mA$

$i_b - i_a = 4i_b$

$i_a = -3i_b$

$i_b = 0.1\overline{6}mA$

$-0.5mA \cdot R + 10k\Omega \cdot i_a + 5k\Omega \cdot i_b + 25V = 0$

$-0.5mA \cdot R + 10k\Omega(-0.5mA) + 5k\Omega \cdot 0.1\overline{6}mA = -25V$

$-0.5mA \cdot R - 5V + 0.83V = -25V$

$\rightarrow R = 41.67\, k\Omega$

Problem 4.7-15

Determine the values of the mesh currents i_1 and i_2 for the circuit shown in Figure P 4.7-15.

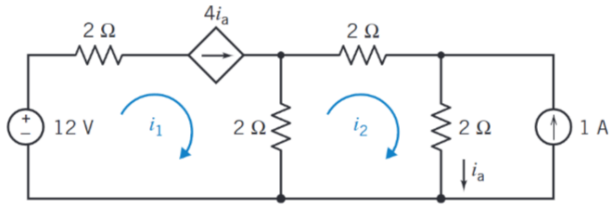

Figure P 4.7-15

$$-12V + 2i_1 + 2(i_1 - i_2) = 0$$

$$2(i_2 - i_1) + 2i_2 + 2(i_2 + 1) = 0$$

$i_1 = 4i_a$ $\qquad i_1 = -8\text{ A}$

$i_2 = i_a - 1$ $\qquad i_2 = -3A$

$$2(i_2 - i_1) + 2i_2 + 2(i_2 + 1) = 0$$

$$6i_2 - 2i_1 + 2 = 0$$

$$6(i_a - 1) - 2(4i_a) + 2 = 0$$

$$-2i_a - 4 = 0 \sim i_a = -2A$$

*Problem 4.8-2

The circuit shown in Figure P 4.8-2 has two inputs, v_s and i_s, and one output v_o. The output is related to the inputs by the equation

$$v_o = ai_s + bv_s$$

where a and b are constants to be determined. Determine the values a and b by

(a) writing and solving mesh equations and

(b) writing and solving node equations.

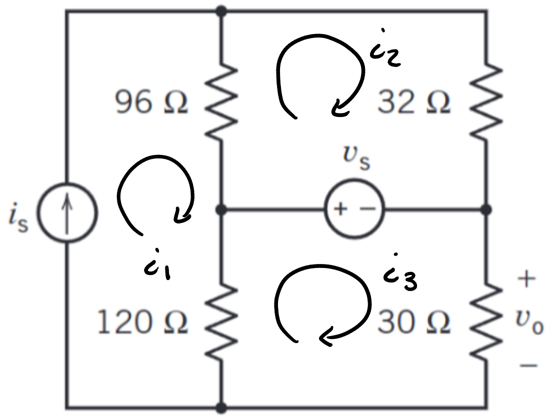

Figure P 4.8-2

$$96(i_s - i_2) + 120(i_s - i_3) = 0$$

$$96(i_2 - i_1) + 32(i_2) - V_s = 0$$

$$120(i_3 - i_s) + V_s + 30i_3 = 0$$

$$96i_s - 96i_2 + 120i_s - 120i_3 = 0$$

$$216i_s - 96i_2 - 120i_3 = 0$$

$$96i_2 = V_s + 64i_s$$

$$i_2 = \frac{V_s + 64i_s}{96}$$

$$96i_2 - 96i_s + 32i_2 - V_s = 0$$

$$128i_2 - 96i_s - V_s = 0$$

$$128\left(\frac{V_s + 64i_s}{96}\right) - 96i_s - V_s = 0$$

$$\frac{4}{3}V_s + \frac{4}{3}(64i_s) - 96i_s - V_s = 0$$

$$V_s/3 - \frac{32}{3}i_s = 0$$

$$V_s - 32i_s = 0 \sim V_s = 32i_s$$

$$120i_3 - 120i_s + V_s + 30i_3 = 0$$

$$150i_3 - 120i_s + V_s = 0$$

$$150i_3 = -V_s + 120i_s$$

$$i_3 = \frac{-V_s}{150} + \frac{4}{5}i_s$$

$a = -1/150$

$b = 4/5$